APPENDIX 13

Geotech Boring Report

Cheyenne VAMC EHRM Infrastructure Upgrades

Geotechnical Engineering Report

November 29, 2023 | Terracon Project No. 24225110

Prepared for:

Specialized Engineering Solutions
10360 Ellison Circle
Omaha, Nebraska 68134

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November 29, 2023

Specialized Engineering Solutions
10360 Ellison Circle
Omaha, Nebraska 68134

Attn: Brad Carne, P.E.
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Re: Geotechnical Engineering Report
Cheyenne VAMC EHRM Infrastructure Upgrades
2360 East Pershing Boulevard
Cheyenne, Wyoming
Terracon Project No. 24225110

Mr. Carne:

We have completed the scope of Geotechnical Engineering services for the Veterans Affairs Medical Center Electronic Health Record Modernization (VAMC EHRM) project in general accordance with Terracon Proposal No. P24225010 dated December 20, 2022. This report presents the findings of the subsurface exploration and provides geotechnical recommendations concerning earthwork and the design and construction of foundations and floor slabs for the proposed project.

We appreciate the opportunity to be of service to you on this project. If you have any questions concerning this report or if we may be of further service, please contact us.

Sincerely,

Terracon

Professional Engineer (Civil)
ERIC D. BERNHARDT
11167
Date 11/29/2023
WYOMING

Kurt F. Stauder, P.G.
Project Geologist

Eric D. Bernhardt, P.E.
Regional Geotechnical Manager

Table of Contents

Figures

GeoModel

Attachments

Exploration and Testing Procedures
Site Location and Exploration Plans
Exploration and Laboratory Results
Supporting Information

Note: This report was originally delivered in a web-based format. **Blue Bold** text in the report indicates a referenced section heading. The PDF version also includes hyperlinks which direct the reader to that section and clicking on the Terracon logo will bring you back to this page. For more interactive features, please view your project online at **client.terracon.com**.

Refer to each individual Attachment for a listing of contents.

Report Summary

Topic [1]	Overview Statement [2]
Project Description	The project will include the construction of a new building addition to the north side of Main Building Cowboy located within the Cheyenne Veterans Affairs Medical Center (VAMC) complex.
Geotechnical Characterization	Some areas of existing fill varying from about 3½ to 5½ feet deep Below the fill, sand with varying amounts of fines were observed to the final depth explored of about 20½ feet. No groundwater was observed. No bedrock was observed
Earthwork	Remove existing fill below the proposed building addition and replace with engineered fill. Where excavations for the proposed addition approach adjacent existing structures with basement areas, CLSM should be used to backfill around these new below-grade foundations and below the entire building addition footprint to at least 5 feet below the bottom of the proposed building addition floor slab elevation. On-site soils or approved imported fill may be used for the remaining fill.
Shallow Foundations	Shallow foundations are recommended for building addition support. Allowable bearing pressure = 2,000 psf for footings adjacent to existing foundations and for stepped footings approaching upper levels; 3,000 psf in areas of new foundations founded on at least 4 feet of newly placed properly compacted fill Expected settlements: <1 inch total, <¾ inch differential
Deep Foundations	Helical piles may also be used for support for the proposed building addition.
Below-Grade Structures	None are expected; however, new building addition foundation planned near basement levels of the existing building adjacent to the proposed addition will extend up to about 12 feet below grade and will be backfill (exterior and interior) to ground/upper level.
Pavements	No pavement construction is anticipated as part of this project.
General Comments	This section contains important information about the limitations of this geotechnical engineering report.

Topic [1]	Overview Statement [2]

1. If the reader is reviewing this report as a pdf, the topics above can be used to access the appropriate section of the report by simply clicking on the topic itself.
2. This summary is for convenience only. It should be used in conjunction with the entire report for design purposes.

Introduction

This report presents the results of our subsurface exploration and Geotechnical Engineering services performed for the proposed building addition to be located at 2360 East Pershing Boulevard in Cheyenne, Wyoming. The purpose of these services was to provide information and geotechnical engineering recommendations relative to:

- Subsurface soil conditions
- Groundwater conditions
- Seismic site classification per IBC
- Site preparation and earthwork
- Demolition considerations
- Foundation design and construction
- Floor slab design and construction

The geotechnical engineering Scope of Services for this project included the advancement of test borings, laboratory testing, engineering analysis, and preparation of this report.

Drawings showing the site and boring locations are shown on the **Site Location** and **Exploration Plan**, respectively. The results of the laboratory testing performed on soil samples obtained from the site during our field exploration are included on the boring logs and/or as separate graphs in the **Exploration Results** section.

Project Description

Our initial understanding of the project was provided in our proposal and was discussed during project planning. A period of collaboration has transpired since the project was initiated, and our final understanding of the project conditions is as follows:

<table>
<tr><th>Item</th><th>Description</th></tr>
<tr><td>Information Provided</td><td>This report is based on information provided by Specialized Engineering Solutions via email correspondences, phone conversations, and a project kick-off conference call beginning on November 29, 2022.
The following documents were referenced during development of this report.
■ Email correspondence containing historic building plans, August 23, 2024 (from Calibre Engineering)
■ Design team meeting, August 22, 2023
■ Terracon's Geotechnical Engineering Report for the Cheyenne VAMC Addition - Terracon Project No. 24215057, dated January 26, 2023 (revised)
■ Architect-Engineer Design Scope of Work for the planned EHRM Infrastructure Upgrades, dated October 25, 2022.
■ Project team collaboration meeting via Microsoft Teams on August 22, 2023</td></tr>
<tr><td>Project Description</td><td>The proposed project includes the construction of a new building addition to the north side of Main Building Cowboy located within the Cheyenne Veterans Affairs Medical Center (VAMC) complex.</td></tr>
<tr><td>Proposed Structure</td><td>We understand the proposed addition will consist of a new at-grade data center (non-basement). The new addition is proposed to be a single story structure and approximately 42 feet long by 24 feet wide; the new floor elevation is planned about 3 feet above existing grade.</td></tr>
<tr><td>Building Construction</td><td>We understand the addition will consist of masonry construction supported on shallow, spread footing or helical pile foundations with a reinforced concrete slab-on-grade floor system.</td></tr>
<tr><td>Finished Floor Elevation FFE)</td><td>Finished floor elevation of the addition is planned to generally match those of the existing building.</td></tr>
<tr><td>Assumed Maximum Loads</td><td>■ Columns: No columns planned
■ Walls: 5 kips per linear foot (klf)
■ Slabs: 150 pounds per square foot (psf)</td></tr>
<tr><td>Grading</td><td>We understand approximately 3 feet of fill is planned for construction of the new building addition in order to match FFE of the existing building.</td></tr>
<tr><td>Pavements</td><td>Not reported as part of site improvements.</td></tr>
</table>

Terracon should be notified if any of the above information is inconsistent with the planned construction, especially the grading limits, as modifications to our recommendations may be necessary.

Site Conditions

The following description of site conditions is derived from our site visit in association with the field exploration and our review of publicly available geologic and topographic maps.

Item	Description
Location	The project is located at 2360 East Pershing Boulevard in Cheyenne, Wyoming. Approximate coordinates near the center of the site are 41.1484° N, 104.7853° W. See **Site Location**.
Existing Improvements	The project site is located at the existing VAMC facility complex on the north side of Main Building Cowboy, between Building 1-B and Building 1-CA Buffalo. The general area includes parking areas/drive lanes, landscaped areas, and concrete flatwork associated with the VAMC complex. Additionally, we understand the existing, adjacent buildings have basement levels that may vary slightly in floor elevations and extend to assumed depths of about 8 to 12 feet below existing building FFE.
Current Ground Cover	Ground cover in the area of the planned building addition consists of landscaping areas and concrete flatwork.
Existing Topography	The site is relatively flat; however, the existing building FFE is about 3 feet above the existing grade at our boring locations.

Geotechnical Characterization

GeoModel

We have developed a general characterization of the subsurface conditions based upon our review of the subsurface exploration, laboratory data, geologic setting, and our understanding of the project. This characterization, termed GeoModel, forms the basis of our geotechnical calculations and evaluation of the site. Conditions observed at each exploration point are indicated on the individual logs. The individual logs can be found in the **Exploration Results** and the GeoModel can be found in the **Figures** attachment of this report.

As part of our analyses, we identified the following model layers within the subsurface profile. For a more detailed view of the model layer depths at each boring location, refer to the GeoModel.

Model Layer	Layer Name	General Description
1	Vegetative Soil	Generally, 6 inches or less in thickness, containing frequent fine root matter
2	Existing Fill	Sand with varying amounts of clay, silt, and gravel. Fine to coarse grained, dark brown and brown
3	Non-Plastic Native Sand	Medium dense to very dense sand with varying amounts of silt and gravel. Fine to coarse grained, light brown and brown.
4	Low-Plasticity Native Sand	Loose to medium dense with varying amounts of silt and clay. Fine to medium grained, light brown to tan.

Groundwater

The boreholes were observed while drilling and shortly after completion for the presence and level of groundwater. Groundwater was not observed in the borings while drilling, or for the short duration the borings were allowed to remain open.

Groundwater levels can and should be expected to fluctuate in response to site development and with varying seasonal and weather conditions and other factors not evident at the time the borings were performed. While not likely, it is possible groundwater may be present during construction or at other times in the life of the facility. Although we do not believe groundwater will impact the proposed construction, surface water should be taken into consideration, and proper site drainage should be implemented in design and construction

Corrosivity

The values presented in the table below may be used to estimate potential corrosive characteristics of the on-site soils with respect to contact with the various underground materials which will be used for project construction. For preliminary recommendations related to these results see **Preliminary Corrosion Recommendations** section.

Corrosivity Test Results Summary									
Boring	**Sample Depth (feet)**	**Soil Description**	**Sulfides (mg/kg)**	**Red-Ox Potential (mV)**	**Soluble Sulfate (mg/kg)**	**Chlorides (mg/kg)**	**Electrical Resistivity[1] (Ω-cm)**	**pH**	**Total Salts (mg/kg)**
B-2	1 to 6	Silty/Clayey sand	0.4	+214	Nil	26	4,550	7.4	2,847

1. Test performed on saturated soil sample.

Seismic Site Class

Seismic design requirements for buildings and other structures are based on Seismic Design Category. Site Classification is required to determine the Seismic Design Category for a structure. The Site Classification is based on the upper 100 feet of the site profile defined by a weighted average value of either shear wave velocity, standard penetration resistance, or undrained shear strength in accordance with Section 20.4 of ASCE 7 and the International Building Code (IBC). Based on the soil/bedrock properties observed at the site and as described on the exploration log and results, we recommend a **Seismic Site Classification of Default D** be considered for the project. Subsurface explorations at this site were extended to a maximum depth of about 20½ feet. The site properties below the boring depth to 100 feet were estimated based on our experience and knowledge of geologic conditions of the general area. Additional deeper borings or geophysical testing may be performed to confirm the conditions below the current boring depth.

Geotechnical Overview

Based on geotechnical conditions encountered in our test borings, the site appears suitable for the proposed construction from a geotechnical point of view provided certain precautions, design, and construction recommendations presented in this report are followed. We have identified some geotechnical conditions that could impact design, construction, and performance of the building and other site improvements. These include the presence of undocumented fill and layers of loose native sand soils within approximately 8 to 13 feet in the proposed building addition area. These conditions will require particular attention in project planning, design and during construction and are discussed in greater detail in the following sections.

Undocumented Fill

Undocumented fill was encountered to depths of approximately 3½ to 5½ feet below the existing ground surface at the time of drilling. We anticipate much deeper fills adjacent to the existing buildings with basement areas associated with backfilling of the basement areas after construction. Undocumented fill could exist at other locations on the site and extend to greater depths. The undocumented fill is assumed to have been placed during the construction of the adjacent existing building; however, we do not possess any definitive information regarding whether the fill was placed and documented under the observation of a geotechnical engineer. Undocumented fill can present a greater than normal risk of post-construction movement of foundations, slabs, and other site improvements supported on or above these materials. Consequently, it is our opinion undocumented fill on the site should not be relied upon for support of foundations and floor slabs for the building addition, and should be removed to native soil, moisture conditioned and recompacted prior to new/engineered fill placement and/or construction.

Where excavations for the removal of existing fills approach or extend below the below existing building foundations, underpinning or excavation bracing and/or shoring will be needed to reduce the risk of undermining the existing building foundations. Existing structure foundation plans should be reviewed prior to design and construction of the planned building addition to evaluate contingencies for alternative excavation methods.

Loose Sands

Loose sand soils were encountered to depths of approximately 2 to 15 feet below existing ground surface within all borings completed at this site. Particularly, the materials were encountered within approximately 8 to 13 feet below ground surface in the proposed building addition area. These materials present a risk for potential settlement of shallow foundations and floor slabs. Additionally, loose native sand soils can be susceptible to disturbance and loss of strength under repeated construction traffic loads and unstable conditions could develop. Stabilization of loose sands may be required at some locations to provide adequate support for construction equipment and proposed structures. Terracon should be contacted if these conditions are encountered to observe the conditions exposed and to provide guidance regarding stabilization (if needed).

Foundation and Floor System Recommendations

The proposed building addition can be supported by a shallow, spread footing or helical pile foundation system and a concrete slab-on-grade floor system can be used for the proposed building addition provided the undocumented fill soils are removed and replaced with moisture conditioned, properly compacted engineered fill and prepared in accordance with the recommendations presented in **Earthwork**. Design recommendations for shallow foundations for the proposed building addition are presented in the **Shallow Foundation** section. We recommend the top 6 inches of backfill below the floor slab consist of WYDOT Grading L or W aggregate base. Design recommendations for slab-on-grade floor systems are presented in the **Floor Slab** section.

Alternatively, consideration be given to supporting the new building addition on helical pile foundations. We anticipate helical piles could be installed more efficiently than spread footings, as removal/replacement of existing fill and risk of underpinning existing shallow foundations are not anticipated for construction of helical piles. Based on the subsurface conditions encountered at our boring locations on this site and the assumed foundation loads, helical pile foundations bearing in native soil should be suitable for support of the proposed building addition. Design recommendations for helical piles are presented in the **Deep Foundations – Helical Piles** section.

The recommendations contained in this report are based upon the results of field and laboratory testing (presented in the **Exploration Results**), engineering analyses, and our current understanding of the proposed project. The **General Comments** section provides an understanding of the report limitations.

Earthwork

The following presents recommendations for demolition, site preparation, excavation, subgrade preparation and placement of engineered fills on the project. Earthwork on the project should be observed and evaluated by Terracon. The evaluation of earthwork should include observation and testing of engineered fill, subgrade preparation, foundation bearing soils, and other geotechnical conditions exposed during the construction of the project.

Demolition

Demolition should include complete removal of existing landscaping features, concrete flatwork, and other improvements within the proposed construction area. This should include removal of any utilities to be abandoned along with any loose utility trench backfill or loose backfill found adjacent to existing foundations

Site Preparation

Site preparation should commence with removal of existing vegetative soils, concrete flatwork, undocumented fill and any loose, soft, or otherwise unsuitable material from the proposed construction areas. Materials containing organics (if any) should be wasted from the site or reused in planned landscaped areas.

After the existing undocumented fill has been removed from within the construction area, the top 10 inches of the exposed ground surface should be scarified, moisture conditioned, and compacted before any new fill is placed. Excavations near existing foundations will likely be in native soils beneath the undocumented fill. The subgrade should then be proof rolled with adequately loaded equipment. The proof rolling should be performed under the observation of the Geotechnical Engineer or representative. Areas excessively deflecting under the proof roll should be delineated and subsequently addressed by the Geotechnical Engineer. Excessively wet or dry material should either be removed, or moisture conditioned and recompacted.

After exposed subgrades and bottoms of excavations have been prepared as recommended above, CLSM and/or engineered fill can be placed to bring the structure pad to the desired grade. Engineered fill should be placed in accordance with the recommendations presented in subsequent sections of this report.

The stability of the subgrade may be affected by precipitation, repetitive construction traffic, or other factors. If unstable conditions develop, workability may be improved by scarifying and drying. Alternatively, over-excavation of wet zones and replacement with granular materials may be used, or crushed gravel and/or rock can be tracked or "crowded" into the unstable surface soil until a stable working surface is attained. Use of geosynthetics could also be considered as a stabilization technique. Lightweight excavation equipment may also be used to reduce subgrade pumping.

Fill Material Types

Fill for this project should consist of engineered fill. Engineered fill is fill that meets the criteria presented in this report and has been properly documented. Existing fill and native onsite soils or approved granular and low plasticity cohesive imported materials should be used as engineered fill material. The earthwork contractor should expect mechanical processing and moisture conditioning of the site soils will be needed to achieve proper compaction.

Imported materials meeting the properties presented below should be acceptable for use as engineered fill. However, imported soils should be evaluated and approved by the geotechnical engineer prior to delivery to the site.

Gradation/Property	Percent Finer by Weight (ASTM C136/D422)
3-inch	100
No. 4 Sieve	30 to 100
No. 200 Sieve	50 (max.)
Soil Properties	**Values**
Liquid Limit (LL)	35 (max.)
Plasticity Index (PI)	15 (max.)

Other import fill material types may be suitable for use depending upon proposed application and location on the site and should be tested and approved for use on a case-by-case basis.

Fill Placement and Compaction Requirements

Placement of engineered fill should be performed according to the following compaction requirements.

Item	Engineered Fill
Maximum Lift Thickness	■ 9 inches or less in loose thickness when heavy, self-propelled compaction equipment is used ■ 4 to 6 inches in loose thickness when hand-guided equipment (i.e. jumping jack or plate compactor) is used
Minimum Compaction Requirements [1,2,3]	■ At least 95% of the standard Proctor maximum dry density (ASTM D698)
Water Content Range [1, 4]	■ Cohesionless soils (sand) [5]: -2% to +2%

1. Engineered fill should be placed and compacted in horizontal lifts, using equipment and procedures that will produce recommended moisture contents and densities throughout the lift. A construction disc or other suitable processing equipment will be needed to thoroughly process the materials and to aid in achieving uniform moisture content throughout the fill.
2. The contractor should expect some moisture adjustment and processing of the site soils or import materials will be needed prior to or during compaction operations.
3. Care should be taken during the fill placement process to avoid zones of dissimilar fill. Improvements constructed over varying fill types are at a higher risk of differential movement compared to improvements over a uniform fill zone.
4. Percent of the optimum moisture content as determined by the standard Proctor test.
5. Specifically, moisture levels should be maintained low enough to allow for satisfactory compaction to be achieved without the fill material pumping when proof rolled.

Special Backfill Considerations

In areas between existing foundation elements as well as areas where the addition of new fill (to accomplish final grade elevation) and in areas where the fill thickness is greater than 5 feet, Controlled Low-Strength Material (CLSM) is recommended as over-excavation backfill. CLSM, also known as flowable fill, will reduce the risk of differential settlement in these areas of comparatively deep fills.

CLSM may need to be placed in multiple lifts to avoid applying significant lateral earth pressures to walls being backfilled. We recommend the structural engineer select acceptable CLSM lift thicknesses. Once the CLSM has achieved about 50 percent strength, the lateral earth pressures will no longer be applied to the walls being backfilled.

Soil backfill placed on top of the CLSM shall not be placed until the CLSM has reached the initial set. Soils placed above the CLSM layer should be compacted to at least 98 percent of the maximum dry density as determined by ASTM D698.

Excavation and Utility Trench Construction

We believe the soils encountered in our exploratory borings can be excavated with conventional excavation equipment. Groundwater seepage is not expected for excavations shallower than about 30½ feet below current site grades. However, if seepage occurs or rain or snow-melt water develops, it should be removed as soon as possible. In order to reduce the potential for subgrade destabilization of the bottom of excavations, consideration should be given to performing excavations remotely with trackhoes or other low ground pressure (LGP) excavation equipment.

Excavations should be made with sufficient working space to permit construction including backfill placement and compaction. Backfill should consist of approved on-site soil or approved import materials. Backfill should be placed and compacted as described under **Compaction Requirements**. It is strongly recommended a representative of the geotechnical engineer provide full-time observation and compaction testing of trench backfill within the building addition.

Utility trenches are a common source of water infiltration and migration. Utility trenches that penetrate beneath the structure should be effectively sealed to restrict water intrusion and flow through the trenches that could migrate below the existing building or proposed addition. We recommend constructing an effective clay “trench plug” that extends at least 5 feet out from the face of the building exterior. The plug material should consist of clay compacted at a water content at or above the soils optimum water content. The clay fill should be placed to completely surround the utility line and be compacted in accordance with recommendations in this report.

Underground piping within or near the proposed structure should be designed and constructed so deviations in alignment do not result in breakage or distress. Utility knockouts in grade beams should be oversized to accommodate differential movements.

Excavation Stability

The individual contractor(s) is responsible for designing and constructing stable, temporary excavations in order to maintain stability of excavation sides and bottom as well as any adjacent structures and foundations. Care should be taken during construction to avoid affecting existing and adjacent structure foundations. The excavations for the project should not undermine existing foundations. Excavations extending below existing adjacent foundations may require shoring of excavations or underpinning of existing foundations to protect the structural integrity of the existing building.

Excavations should be sloped or shored in the interest of safety following local and federal regulations, including current Occupational Safety and Health Administration (OSHA) excavation and trench safety standards. Based on the soils observed in the boring log, the onsite soils generally classify as an OSHA Type C soil. Vertical excavations may extend up to 4 feet in depth. If excavations greater than 4 feet are desired, the ground can be laid back at a slope of 1½:1 (Horizontal: Vertical). As a safety measure, it is recommended vehicles and soil piles be kept to a minimum lateral distance from the crest of the slope equal to no less than the slope height. Exposed slope faces should be protected against the elements.

The soils to be exposed by the proposed excavations may vary significantly across the site. The soil classifications are based solely on the materials encountered in widely spaced exploratory test borings. The contractor should verify similar conditions exist throughout the proposed area of excavation. If different subsurface conditions are encountered at the time of construction, the actual conditions should be evaluated to determine any excavation modifications necessary to maintain safe conditions.

Grading and Drainage

Proper drainage and surface water management is important to the performance of foundations, floor slabs, and other site improvements. The following recommendations are considered good practice for any site and should be implemented where applicable and/or to the extent possible.

Grades must be adjusted to provide positive drainage away from existing and proposed structures as well as proposed site improvements during construction and maintained throughout the life of the proposed facility. Infiltration of water into utility or foundation excavations must be prevented during construction.

Landscaped irrigation adjacent to the foundation system should be minimized. Plants placed close to foundation walls should be limited to those with low moisture requirements. The importance of proper irrigation practices cannot be over emphasized. Irrigation should be limited to the minimum amount needed to maintain vegetation; application of more water will increase likelihood of slab and foundation movements.

The ground surface should be sloped in such a manner that water will not pond between or adjacent to the building addition, existing building, and other site improvements.

Concrete curbs and sidewalks may “dam” surface runoff adjacent to the building and disrupt proper flow. Use of “chase” drains or weep holes at low points in the curb should be considered to promote proper drainage.

Backfill against foundations, exterior grade beams and in utility and sprinkler line trenches should be well compacted and free of construction debris to reduce moisture infiltration. We recommend exterior foundation wall backfill consist of on-site soils or approved import clays to reduce infiltration and conveyance of surface water through the backfill. Some settlement of wall backfill should be expected even if properly compacted. Areas where backfill has settled should be repaired and re-graded immediately to maintain proper slope away from the foundation.

Flatwork and pavements will be subject to post construction movement. Maximum grades practical should be used for paving and flatwork to prevent areas where water can pond. Where paving or flatwork abuts the structure, care should be taken that joints are properly sealed and maintained to prevent the infiltration of surface water.

Planters located adjacent to the structure (if any) should be self-contained. Sprinkler mains and spray heads should not be installed or allowed to discharge within 5 feet of foundation walls. Roof drains should discharge on pavements or be extended away from the structure well beyond the limits of the backfill zone through the use of splash blocks or downspout extensions. Downspouts and extensions should be monitored and maintained in good working condition. Generally speaking, downspouts should not be buried and extended below grade, as these systems can be difficult to monitor and maintain.

Water permitted to pond near or adjacent to the perimeter of the existing structure or proposed addition (either during or post-construction) can result in higher soil movements than those discussed in this report. As a result, estimations of potential movement described in this report cannot be relied upon if positive drainage is not obtained and maintained, and water is allowed to infiltrate the fill and/or subgrade.

After building addition construction and prior to project completion, we recommend verification of final grading be performed to document that positive drainage, as described in this section, has been achieved. Maintenance of surface drainage is imperative subsequent to construction and becomes the responsibility of the owner.

Earthwork Construction Considerations

Unstable subgrade conditions may be anticipated at the base of excavations after removal of existing fill. Should unstable subgrade conditions be encountered, they will need to be corrected prior to the placement of new fill or construction. Terracon should be contacted if these conditions are encountered to observe the exposed conditions and to provide guidance regarding stabilization (if needed). The appropriate method of stabilization should be determined once the entire excavation is performed, and the full extent of the unstable subgrade conditions can be observed.

The subgrade should be evaluated by a Terracon representative upon completion of over-excavation and subsequent backfilling operations. Once backfill is placed and the subgrade is prepared, it is important measures be planned and taken to reduce drying of the near surface materials. If the fill dries excessively prior to grade beam or floor slab construction, then it will be necessary to rework the upper, drier materials just prior to construction. Likewise, completed subgrades that have become saturated, frozen, disturbed, or altered by construction activity should be restored to the conditions recommended in this report.

Construction Observation and Testing

The earthwork efforts should be monitored under the guidance of Terracon. Monitoring should include documentation of over-excavation and reworking/replacement of existing fill soils (as described above), proof-rolling and mitigation of areas delineated by the proof-roll to require mitigation. Each lift of compacted fill should be tested, evaluated, and reworked as necessary until approved by the Geotechnical Engineer prior to placement of additional lifts.

In addition to the documentation of the essential parameters necessary for construction, the continuation of Terracon into the construction phase of the project provides the continuity to maintain our evaluation of subsurface conditions, including assessing variations and associated design changes.

Shallow Foundations

If the site has been prepared in accordance with the requirements noted in **Geotechnical Characterization** and **Earthwork**, the following design parameters are applicable for shallow foundations.

Shallow Foundation – Design Recommendations

Description	Description
Bearing material	Engineered fill and removal of existing fill and prepared in accordance with the **Earthwork** section of this report.

Description	Description	
Maximum net allowable soil bearing pressure [1]	Footings located adjacent to existing footings and stepped footings between basement elevations and upper levels	Footings placed on at least 4 feet of newly placed engineered fill
	2,000 psf	3,000 psf
Minimum embedment below finished grade for frost protection [2]	36 inches	
Estimated post-construction movement based on structural loads provided [3]	About 1 inch of total movement and differential movement on the order of about ½ to ¾ inch	
Ultimate passive pressure [4]	225 psf/ft.	
Ultimate coefficient of sliding friction [4]	0.40 times the vertical dead load	

1. The allowable soil bearing pressure applies to dead loads plus design live load conditions and is the maximum pressure that should be transmitted to the bearing soils in excess of the minimum surrounding overburden pressure at the footing base elevation. Assumes unsuitable bearing soils, if encountered, will be undercut and replaced with engineered fill.
2. For perimeter footings and footings beneath unheated areas. Interior column pads and footings in heated areas should bear at least 18 inches below the adjacent grade (or the top of the floor slab) for confinement of the bearing materials and to develop the recommended bearing pressure.
3. Foundation movement will depend upon variations within the subsurface soil profile, structural loading conditions, embedment depth of footings, thickness of compacted fill, and the quality of the earthwork operations. Settlement estimates are based on the assumed structural loads and resulting foundation geometry. If actual foundation loads vary significantly from those assumed, we should be contacted to review our recommendations. Additional foundation movements could occur if surface water infiltrates the foundation soils; therefore, proper drainage away from the foundation system should be provided in the final design, during construction and maintained throughout the life of the structure.
4. The sides of the excavation for spread footings must be nearly vertical and the concrete should be placed neat against these vertical faces or backfill must be compacted to at least 95 percent of the standard Proctor maximum dry density for the passive earth pressure value to be valid. Passive pressure requires movement to generate the resistance and should only be used when movement is tolerable, and the soil is well compacted and will not be removed. The passive resistance and friction factor are ultimate values. As such, appropriate factors of safety should be applied.

Description	Description

5. Foundation movement will depend upon variations within the subsurface soil profile, structural loading conditions, embedment depth of foundations, thickness of compacted fill, and the quality of the earthwork operations. Settlement estimates are based on the assumed structural loads and foundation geometry as discussed in the **Project Description** section and as indicated in the table above. If the actual foundation loads or geometry vary significantly from those assumed, we should be contacted to review our recommendations. Additional foundation movements could occur if surface water infiltrates the foundation soils; therefore, proper drainage away from the foundation system should be provided in the final design, during construction and maintained throughout the life of the structure.

Excavations for fill extending below the bottom of foundation elevation should extend laterally beyond all edges of the foundation at least 8 inches per foot of fill depth below the foundation base elevation. The excavation should be backfilled to the foundation base elevation in accordance with the recommendations presented in this report.

Shallow Foundation Construction Considerations

Differential settlement between the addition and the existing building is expected to approach the magnitude of the total settlement of the addition. Controlled low-strength materials (CLSM) should be used as over-excavation backfill in areas near existing backfill.

Expansion joints should be provided between the existing building and the proposed addition to accommodate differential movements between the two structures. Underground piping between the two structures should be designed with flexible couplings and utility knockouts in foundation walls should be oversized, so minor deflections in alignment do not result in breakage or distress.

Some of the new footings are planned to bear at or near the bearing elevation of immediately adjacent existing foundations. Depending upon their locations and current loads on the existing footings, footings for the new addition could cause settlement of adjacent walls. To reduce this concern and risk, clear distances at least equal to the width of the new footing should be maintained between the addition's footings and footings supporting the existing building.

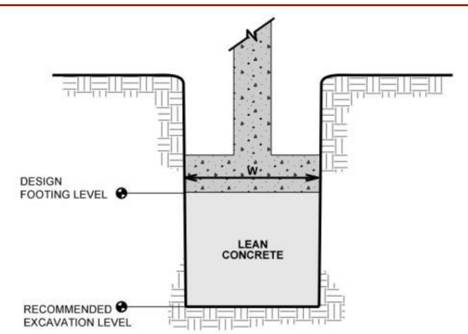
The new footing system is planned to step-up in elevation, from the depth of the existing footings supporting adjacent basement levels for existing structures to approximately frost depth as foundations continue away from existing structures.

Footings should be proportioned to reduce differential foundation movement. Proportioning on the basis of relative constant dead-load pressure can provide a means to reduce differential movement between adjacent footings. Footings should be detailed and reinforced as necessary to reduce the potential for distress caused by differential foundation movement. Additionally, foundations should be constructed on similar materials. Foundations should not be constructed partially on engineered fill and partly on native soil. If placed on engineered fill, it should be at least 18 inches in thickness.

Spread Footing Construction Considerations

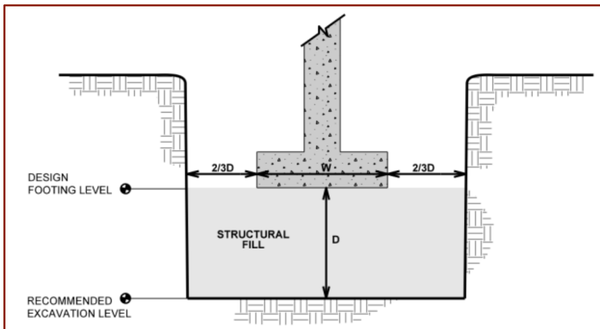
Care should be used while excavating adjacent to foundations of the existing building to avoid undermining or otherwise disturbing the existing foundation bearing soils. Excavations should not extend into the stress influence zone of existing foundations unless they are underpinned or braced. The stress influence zone is defined as the area below a line projected down at a 1H:1V slope from the bottom edge of the existing foundation. If excavations need to extend below the depths of existing foundations, we should be contacted to evaluate conditions and provide additional recommendations, if needed. Also, when compacting within 10 feet of the existing structure, a lightweight compactor should be used and backfill should be placed and compacted in 4- to 6-inch-thick loose lifts.

Where soils are loosened during excavation or in the forming process for footings or if otherwise unsuitable materials are present at foundation bearing depth, these materials should be removed to minimum depths determined by the geotechnical engineer and the resulting excavation should be backfilled up to footing base elevation with either lean concrete or approved fill material placed and compacted as described in the **Earthwork** section of this report. This is illustrated in the sketches below.



LEAN CONCRETE BACKFILL

NOTE: EXCAVATIONS ARE SHOWN VERTICAL; HOWEVER, THE SIDEWALLS SHOULD BE SLOPED AS NECESSARY FOR SAFETY



OVER-EXCAVATION / BACKFILL ZONE

NOTE: EXCAVATIONS ARE SHOWN VERTICAL; HOWEVER, THE SIDEWALLS SHOULD BE SLOPED AS NECESSARY FOR SAFETY

The base of foundation excavations should be free of water and loose soil prior to concrete placement. Concrete should be placed soon after excavating to reduce bearing soil disturbance. Should the soils at bearing level become excessively dry, disturbed or saturated, or frozen, the affected soil should be removed prior to placing concrete.

Completed foundation excavations should be observed and evaluated by a representative of Terracon well in advance of forming footings and placement of reinforcing steel to confirm satisfactory bearing materials are present and subsurface conditions are consistent with those encountered in our borings. If the soil conditions encountered differ significantly from those presented in this report, supplemental recommendations will be required.

Deep Foundations – Helical Piles

Helical Pile Design Recommendations

Based on the subsurface conditions encountered at our boring locations on this site and the anticipated foundation loads, helical pile foundations advanced and bearing in native soil should be suitable for support of the proposed building additions. Design recommendations for foundations for the proposed building addition and related structural elements are presented in the following sections. An experienced helical pile contractor should be consulted to review the Boring Logs to evaluate the use of this foundation system.

The following table lists input values for use in LPILE analyses. Modern versions of LPILE provide estimated default values of k_h and E_{50} based on strength and are recommended for the project. Since deflection or a service limit criterion will most likely control lateral capacity design, no safety/resistance factor is included with the parameters.

Stratigraphy [1]		LPILE Soil Model	ϕ [2]	γ' (pcf) [2]	K_s (pci)
Depth Below Ground Surface (feet)	Material				
0 to 20.5	Silty/Clayey Sand	Sand (Reese)	34	115	90

1. See Subsurface Profile in **Geotechnical Characterization** for more details on Stratigraphy.
2. Definition of Terms:
   ϕ: Internal friction angle
   γ': Effective unit weight
   K_s: Soil Modulus

We do not recommend using vertically installed helical piles to resist lateral loads without approved lateral load test data, as these types of foundations are typically designed to resist axial loads. Only the horizontal component of the allowable axial load should be considered to resist the lateral loading and only in the direction of the batter.

Additionally, Terracon recommends helical piles be installed a minimum of 8 feet into the native soil. Based on our experience, if helical piles are connected to a foundation and the foundation is not attached structurally to the floor slab, the floor slab can experience significantly more settlement than the helical pier supported foundation. Consideration should be given to supporting the floor slab with a grade beam that include helical pile supports.

If a helical pile foundation system is selected by the project team, we recommend the helical pile designer follow the recommendations presented in Chapter 18 of the applicable International Building Code (IBC). We recommend the helices for each helical pile bear in the native, medium dense to dense sand soils encountered below the site, at least 8 feet below existing grade. The helical pile designer should select the size and number of helices for each helical pile based on planned loads and bearing materials described in our exploratory boring logs.

Helical Pile – Construction Considerations

Terracon should be retained to observe helical pile installation to verify proper bearing materials have been encountered during installation and that helical piers achieve the minimum required load carrying ability. Each helical pile should have at least the following information recorded by the contractor during construction; final torque, final depth, materials used, and who was the installer.

In accordance with local building code requirements, a load test should be performed by the helical pile installer to validate achieving the allowable design load. If more than 20 helical piles will be installed, a verification load test should be performed on at least one helical pile using the same materials and installation techniques as production. Testing may be performed in tension (ASTM D1143) or compression (ASTM D3689). The verification test should be completed using a load cell that has been calibrated with one year of the date of the test. If less than 20 helical piles are installed, no verification test is required; however, the structural engineer should use a more conservative factor of safety in the design process.

Torque measurements during installation of helical piles should be used to verify the axial capacity of the helical piles. Terracon should be provided with the torque to capacity relationships for each type of pile used on the project for our review and comment prior to mobilization to the site. We recommend the helical pile installation contractor provide confirmation the installation equipment has been calibrated within one year of installation at this project. The helical foundations should be installed by a qualified specialty contractor per the manufacturer’s recommendations.

We recommend a minimum distance of 15 inches from the edge of the pile caps to the center of each pile. Multiple helical piles within a single pile cap should be spaced apart at least 12 inches or equal to the largest helical bearing plate, whichever is greater. Helical piles are generally spaced several helical bearing plated diameters apart on-center to avoid efficiency effects. Because helical pile shafts are generally several times smaller than the helical bearing plates, their shafts are very far apart relative to their shaft diameter and the lateral capacity of a group of helical piles will rarely be critical in design.

Floor Slabs

Based on our boring data, a concrete slab-on-grade should be a suitable floor system for the proposed building addition, provided the recommendations presented in the **Earthwork** section of this report are followed. This includes removal of all existing fills beneath floor slab and replacing fills with properly moisture conditioned and compacted on-site soils. In areas where fill over-excavation is greater than 5 feet, controlled low-strength materials should be placed. All slabs-on-grade undergo some movement. We believe risk of movement is low for the soil conditions encountered on this site and estimate settlement of slabs-on-grade constructed on properly prepared subgrade to be less than 1 inch. Where slab movement cannot be accepted or must be reduced, we are available to discuss floor movement mitigation techniques upon your request.

Floor Slab Design Parameters

Item	Description
Floor Slab Support [1]	Minimum 6-inches of granular base course (WYDOT Grading L or W) and removal of existing fill.
Estimated Modulus of Subgrade Reaction [2]	For limited area loads or concentrated/point loads placed directly on slabs: ▪ 175 psi/in.
Slab Thickness	Slab reinforcement and thickness should be designed by a qualified engineer based on actual loads imposed and on intended slab use.

1. Modulus of subgrade reaction is an estimated value based upon our experience with the subgrade condition, the requirements noted in **Earthwork**, and the floor slab support as noted in this table. It is provided for point loads. For large area loads, the modulus of subgrade reaction would be lower.

We recommend the following precautions be observed where slabs-on-grade are used. These precautions will not eliminate slab movement, but they tend to reduce damage when movement occurs. Additional floor slab design and construction recommendations are as follows:

- Positive separations and/or isolation joints should be provided between slabs and foundations, columns or utility lines to allow free vertical movement. This detail can reduce cracking when movement of the slab occurs. Non-bearing partition walls placed on the floor slab (if any) should be designed and constructed to allow at least 1 inch of slab movement.

- Frequent control joints should be provided in slabs to control the location and extent of cracking in accordance with the American Concrete Institute (ACI). For additional recommendations refer to the ACI Design Manual.

- The use of a vapor retarder should be considered beneath concrete slabs on grade that will be covered with wood, tile, carpet or other moisture sensitive or impervious coverings, or when the slab will support equipment sensitive to moisture. When conditions warrant the use of a vapor retarder/barrier, the slab designer should refer to ACI 302 and/or ACI 360 for procedures and cautions regarding the use and placement of a vapor retarder/barrier.

- Other design and construction considerations, as outlined in the ACI Design Manual, Section 302.1R are recommended.

Floor Slab Construction Considerations

Fill/backfill placed beneath slabs should be moisture conditioned and compacted as described in the **Earthwork** section of this report. Soils loosened during excavation or other construction activities should be removed or recompacted as described in this report. Floor slabs should not be constructed on frozen subgrade.

Once fill is placed and the subgrade is prepared, it is important measures be planned and taken to reduce drying of the near surface materials. If the fill dries excessively prior to building construction, then it will be necessary to rework the upper, drier materials just prior to installing floor slabs.

On most project sites, site grading is generally accomplished early in the construction phase. However, as construction proceeds, the subgrade may be disturbed due to utility excavations, construction traffic, desiccation, rainfall, etc. As a result, the floor slab subgrade may not be suitable for placement of base and/or concrete and corrective action will be required.

We recommend the area underlying the floor slab be proof rolled shortly before slab construction. Particular attention should be paid to areas where backfilled trenches are located. Areas where unsuitable conditions are located should be repaired by removing and replacing the affected material with properly compacted fill. Floor slab subgrade areas should be moisture conditioned and properly compacted to the recommendations in this report shortly before placement of the base and/or concrete.

Preliminary Corrosion Recommendations

Chemical testing was completed on a near surface sample as part of this investigation. The test results are discussed in the section titled Corrosivity of Earth Materials. Detailed results are presented in **Exploration and Laboratory Results**.

- Site soils are expected to exhibit moderate corrosivity when in direct contact with steel.

- Site soils are expected to exhibit low corrosivity when in direct contact with concrete. As such, conventional Type I Portland cement may be used. However, if there is no, or minimal cost differential, use of Type II Portland cement (or equivalent) should be considered for additional sulfate resistance of construction concrete.

General Comments

Our analysis and opinions are based upon our understanding of the project, the geotechnical conditions in the area, and the data obtained from our site exploration. Variations will occur between exploration point locations or due to the modifying effects of construction or weather. The nature and extent of such variations may not become evident until during or after construction. Terracon should be retained as the Geotechnical Engineer, where noted in this report, to provide observation and testing services during pertinent construction phases. If variations appear, we can provide further evaluation and supplemental recommendations. If variations are noted in the absence of our observation and testing services on-site, we should be immediately notified so that we can provide evaluation and supplemental recommendations.

Our Scope of Services does not include either specifically or by implication any environmental or biological (e.g., mold, fungi, bacteria) assessment of the site or identification or prevention of pollutants, hazardous materials or conditions. If the owner is concerned about the potential for such contamination or pollution, other studies should be undertaken.

Our services and any correspondence are intended for the sole benefit and exclusive use of our client for specific application to the project discussed and are accomplished in accordance with generally accepted geotechnical engineering practices with no third-party beneficiaries intended. Any third-party access to services or correspondence is solely for information purposes to support the services provided by Terracon to our client. Reliance upon the services and any work product is limited to our client and is not intended for third parties. Any use or reliance of the provided information by third parties is done solely at their own risk. No warranties, either express or implied, are intended or made.

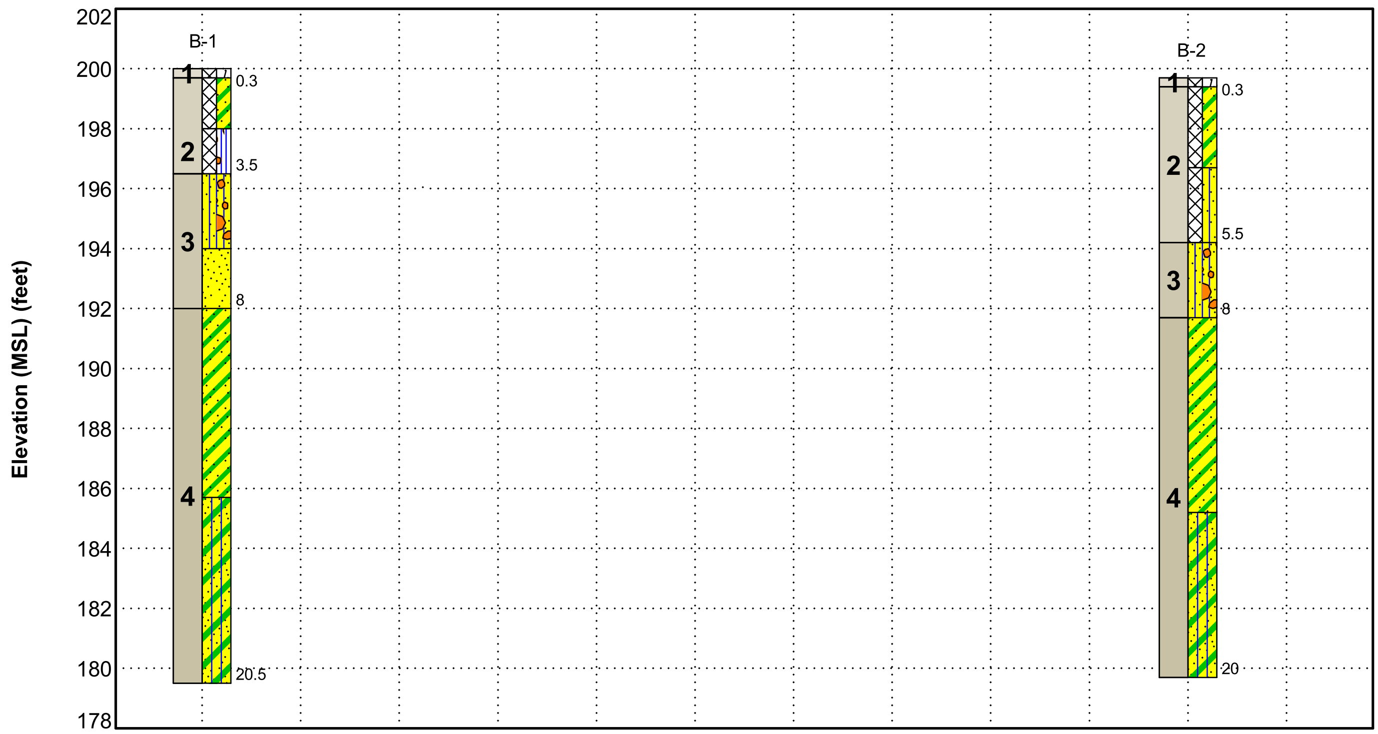
Site characteristics as provided are for design purposes and not to estimate excavation cost. Any use of our report in that regard is done at the sole risk of the excavating cost estimator as there may be variations on the site that are not apparent in the data that could significantly affect excavation cost. Any parties charged with estimating excavation costs should seek their own site characterization for specific purposes to obtain the specific level of detail necessary for costing. Site safety and cost estimating including excavation support and dewatering requirements/design are the responsibility of others. Construction and site development have the potential to affect adjacent properties. Such impacts can include damages due to vibration, modification of groundwater/surface water flow during construction, foundation movement due to undermining or subsidence from excavation, as well as noise or air quality concerns. Evaluation of these items on nearby properties are commonly associated with contractor means and methods and are not addressed in this report. The owner and contractor should consider a preconstruction/precondition survey of surrounding development. If changes in the nature, design, or location of the project are planned, our conclusions and recommendations shall not be considered valid unless we review the changes and either verify or modify our conclusions in writing.

Figures

GeoModel

Terracon
1505 Old Happy Jack Rd
Cheyenne, WY

GeoModel



This is not a cross section. This is intended to display the Geotechnical Model only. See individual logs for more detailed conditions.

Model Layer	Layer Name	General Description
1	**Vegetative soil**	About 3 inches of root penetration.
2	**Existing Fill**	Sand with varying amounts of clay, silt, and gravel. Fine to coarse grained, dark brown and brown.
3	**Non-Plastic Native Sand**	Medium dense to very dense sand with varying amounts of silt and gravel. Fine to coarse grained, light brown and brown.
4	**Low-Plasticity Native Sand**	Loose to medium dense with varying amounts of silt and clay. Fine to medium grained, light brown to tan.

LEGEND

- Vegetative Soil
- Clayey Sand
- Well-graded Sand with Silt and Gravel
- Silty Sand with Gravel
- Poorly-graded Sand
- Silty Clayey Sand
- Silty Sand

NOTES:
Layering shown on this figure has been developed by the geotechnical engineer for purposes of modeling the subsurface conditions as required for the subsequent geotechnical engineering for this project. Numbers adjacent to soil column indicate depth below ground surface.

Attachments

Exploration and Testing Procedures
Site Location and Exploration Plans
Exploration and Laboratory Results
Supporting Information

Exploration and Testing Procedures

Field Exploration

Number of Borings	Approximate Boring Depth (feet)	Location
2	About 20	Planned addition area

Boring Layout and Elevations: Terracon personnel provided the boring layout using handheld GPS equipment (estimated horizontal accuracy of about ±10 feet) and referencing existing site features. Approximate ground surface elevations were obtained by the field exploration crew using a level and grade rod referenced to a temporary benchmark... If surface elevations and a more precise boring layout are desired, we recommend borings be surveyed.

Subsurface Exploration Procedures: We advanced the borings with a truck-mounted drill rig using solid-stem, continuous-flight augers. Five samples were obtained in the upper 10 feet of each boring and at intervals of 5 feet thereafter. Soil sampling was performed using modified California barrel and/or split-barrel sampling procedures. A bulk sample was collected from the cuttings in each boring from about 1 to 6 feet below existing grade. In the split-barrel sampling procedure, a standard 2-inch outer diameter split-barrel sampling spoon was driven into the ground by a 140-pound automatic hammer falling a distance of 30 inches. The number of blows required to advance the sampling spoon the last 12 inches of a normal 18-inch penetration is recorded as the Standard Penetration Test (SPT) resistance value. Modified California barrel sampling procedures are similar to standard split spoon sampling procedure; however, blow counts are typically recorded for 6-inch intervals for a total of 12 inches of penetration. The SPT resistance values, also referred to as N-values, are indicated on the boring logs at the test depths. For safety purposes, all borings were backfilled with auger cuttings after their completion

We also observed the boreholes while drilling and at the completion of drilling for the presence of groundwater. Groundwater was not observed at these times in the boreholes.

The sampling depths, penetration distances, and other sampling information was recorded on the field boring logs. The samples were placed in appropriate containers and taken to our soil laboratory for testing and classification. Our exploration team prepared field boring logs as part of the drilling operations. These field logs included visual classifications of the materials observed during drilling and our interpretation of the subsurface conditions between samples. Final boring logs were prepared from the field logs. The final boring logs represent our interpretation of the field logs and include modifications based on observations and tests of the samples in our laboratory.

Laboratory Testing

We reviewed the field data and assigned laboratory tests. The laboratory testing program included the following types of tests:

- Moisture Content
- Dry Unit Weight
- Atterberg Limits
- Grain Size Distribution
- Corrosion Suite (pH, soluble sulfates, sulfides, chlorides, RedOx, total salts, and electrical resistivity)

The laboratory testing program included examination of soil samples by an engineer. Based on the results of our field and laboratory programs, we described and classified the soil samples in accordance with the Unified Soil Classification System.

Site Location and Exploration Plans

Contents:

Note: All attachments are one page unless noted above.

Site Location

DIAGRAM IS FOR GENERAL LOCATION ONLY, AND IS NOT INTENDED FOR CONSTRUCTION PURPOSES

MAP PROVIDED BY MICROSOFT BING MAPS

Exploration Plan

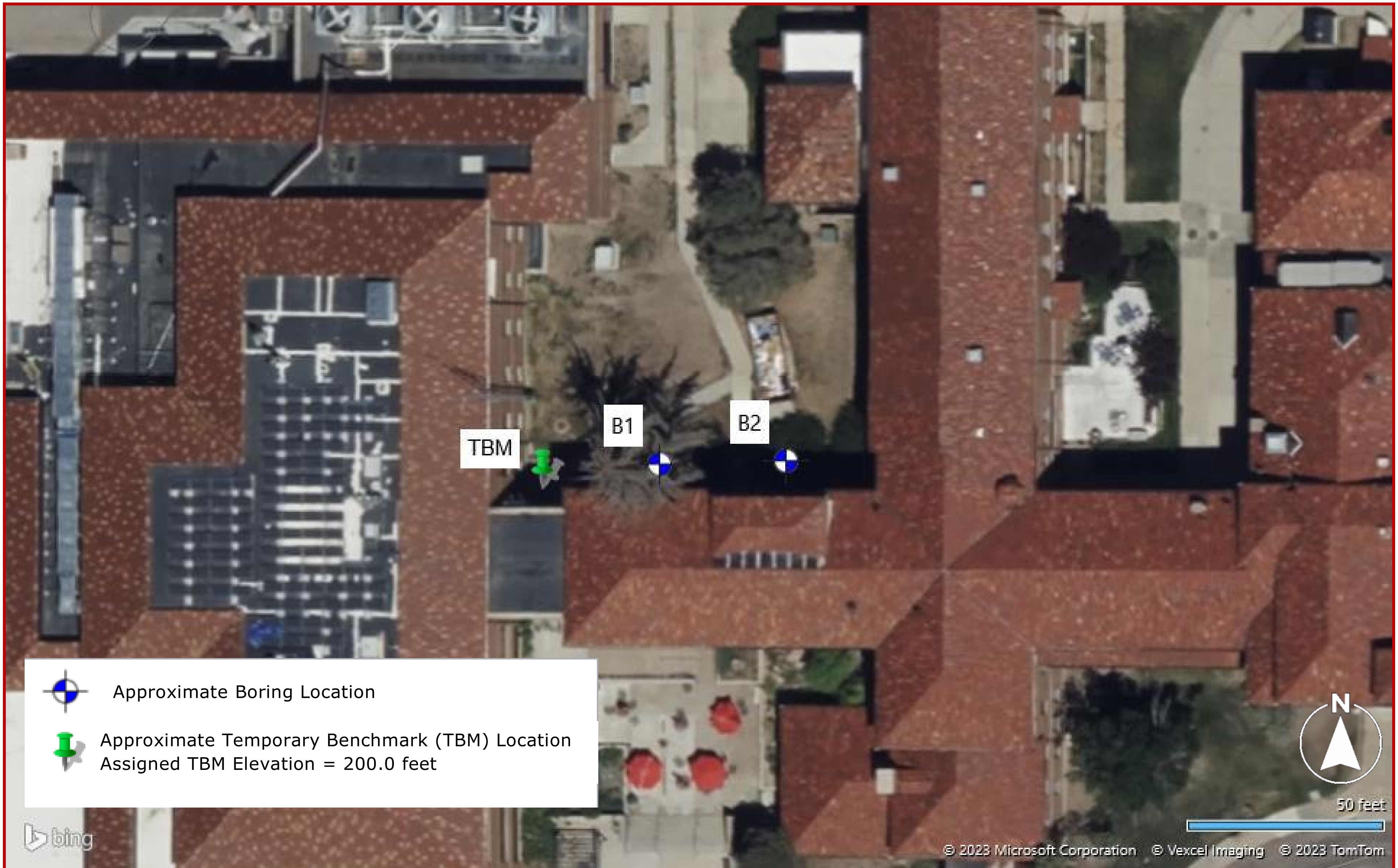


DIAGRAM IS FOR GENERAL LOCATION ONLY, AND IS NOT INTENDED FOR CONSTRUCTION PURPOSES

MAP PROVIDED BY MICROSOFT BING MAPS

Exploration and Laboratory Results

Contents:

Note: All attachments are one page unless noted above.

Terracon
1505 Old Happy Jack Rd
Cheyenne, WY

Boring Log B-1

Location: See Exploration Plan

Latitude: 41.1484° Longitude: -104.7857°

Depth (Ft.)	Description	Elevation: 200.0 (Ft.) +/-	Water Level Observations	Field Test Results	Water Content (%)	Dry Unit Weight (pcf)	Atterberg Limits LL-PL-PI	Percent Fines
0.3	**FILL - VEGETATIVE SOIL**, about 3 inches of root penetration	199.7						
2.0	**FILL - CLAYEY SAND (SC)**, trace fine grained gravel, fine to medium grained sand, dark brown	198		5-5-4-4 N=9	8.9			
3.5	**FILL - WELL GRADED SAND WITH SILT AND GRAVEL (SW-SM)**, fine to coarse grained, brown	196.5		4-4-10-17 N=14	3.8		NP	7
6.0	**SILTY SAND WITH GRAVEL (SM)**, fine grained gravel, fine to coarse grained sand, brown, very dense	194		12-26-40-43 N=66	7.9		NP	18
8.0	**POORLY GRADED SAND (SP)**, trace fine grained gravel, fine to coarse grained sand, light brown, dense	192		13-21-18-14 N=39	2.0			
14.3	**CLAYEY SAND (SC)**, fine to medium grained, brown, loose	185.7		7-5-4-8 N=9	10.4			
				21/12"	18.8	100		
20.5	**SILTY CLAYEY SAND (SC-SM)**, fine to medium grained, tan, medium dense	179.5		5-11-11 N=22	19.2			

Boring Terminated at 20.5 Feet

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Water Level Observations
Groundwater not encountered

Drill Rig
CME 75

Hammer Type
Automatic

Driller
Terracon Consultants, Inc.

Notes

Elevation Reference: Elevations were measured in the field using an engineer's level and grade rod referencing base of the entrance stairway as a Temporary Benchmark (TBM). Assigned TBM Elevation = 200.0 feet. See Exploration Plan for TBM location.

Advancement Method
4-inch diameter, solid-stem auger

Abandonment Method
Boring backfilled with auger cuttings upon completion.

Boring Started
07-06-2023

Boring Completed
07-06-2023

Terracon
1505 Old Happy Jack Rd
Cheyenne, WY

Boring Log B-2

Location: See Exploration Plan

Latitude: 41.1484° Longitude: -104.7856°

Elevation: 199.7 (Ft.) +/-

Depth (Ft.)	Material Description	Elevation (Ft.)	Sample Depth (Ft.)	Field Test Results	Water Content (%)	Dry Unit Weight (pcf)	Atterberg Limits LL-PL-PI	Percent Fines
0.3	**FILL - VEGETATIVE SOIL**, about 3 inches of root penetration	199.4						
3.0	**FILL - CLAYEY SAND (SC)**, trace fine grained gravel, fine to medium grained sand, dark brown	196.7	0–2	4-4-2-3 N=6	11.7			
			2–4	4-11-13-16 N=24	22.2			
5.5	**FILL - SILTY SAND (SM)**, trace fine grained gravel, fine to coarse grained sand, brown	194.2	4–6	14-15-16-18 N=31	1.8			
8.0	**SILTY SAND WITH GRAVEL (SM)**, fine grained gravel, fine to coarse grained sand, brown to light brown, medium dense	191.7	6–7.5	12-11-17 N=28	8.7		NP	17
14.5	**CLAYEY SAND (SC)**, fine to medium grained, light brown, loose to medium dense	185.2	8–10	4-4-4-4 N=8	17.2			
			14–15	18/12"	13.0	102		
20.0	**SILTY CLAYEY SAND (SC-SM)**, fine to medium grained, tan, medium dense	179.7	19–20	24/12"	21.0	108		

Boring Terminated at 20 Feet

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Elevation Reference: Elevations were measured in the field using an engineer's level and grade rod referencing base of the entrance stairway as a Temporary Benchmark (TBM). Assigned TBM Elevation = 200.0 feet. See Exploration Plan for TBM location.

Water Level Observations
Groundwater not encountered

Advancement Method
4-inch diameter, solid-stem auger

Abandonment Method
Boring backfilled with auger cuttings upon completion.

Drill Rig
CME 75

Hammer Type
Automatic

Driller
Terracon Consultants, Inc.

Boring Started
07-06-2023

Boring Completed
07-06-2023

Terracon
1505 Old Happy Jack Rd
Cheyenne, WY

Boring Log No. B-3

Location: See Exploration Plan

Latitude: 41.1497° Longitude: -104.7855°

Elevation: 694 (Ft.) +/-

Depth (Ft.)	Material Description	Elevation (Ft.)	Sample Depth (Ft.)	Field Test Results	Water Content (%)	Dry Unit Weight (pcf)	Atterberg Limits LL-PL-PI	Percent Fines
0.6	**VEGETATIVE SOIL**, about 7 inches of root penetration	693.4		4-6-8 N=14				
3.5	**SILTY CLAYEY SAND (ML)**, trace fine grained gravel, fine to medium grained sand, tan, medium dense	690.5		34/12"				
14.0	**POORLY GRADED SAND WITH GRAVEL (SP)**, fine grained gravel, fine to coarse grained, orange to tan, medium dense to dense light reddish gray at about 6 feet	680		9-19-20 N=39 41/12"				
20.5	**SILTY CLAYEY SAND (SC-SM)**, trace fine grained sand, tan, very stiff	673.5		28/12" 7-11-14 N=25				

Boring Terminated at 20.5 Feet

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Elevation Reference: Elevations were measured in the field using an engineer's level and grade rod referencing base of the entrance stairway as a Temporary Benchmark (TBM). Assigned TBM Elevation = 200.0 feet. See Exploration Plan for TBM location.

Water Level Observations
Groundwater not encountered

Advancement Method
4-inch diameter, solid-stem auger

Abandonment Method
Boring backfilled with auger cuttings upon completion.

Drill Rig
CME 75

Hammer Type
Automatic
Efficiency 71%

Driller
Terracon Consultants, Inc.

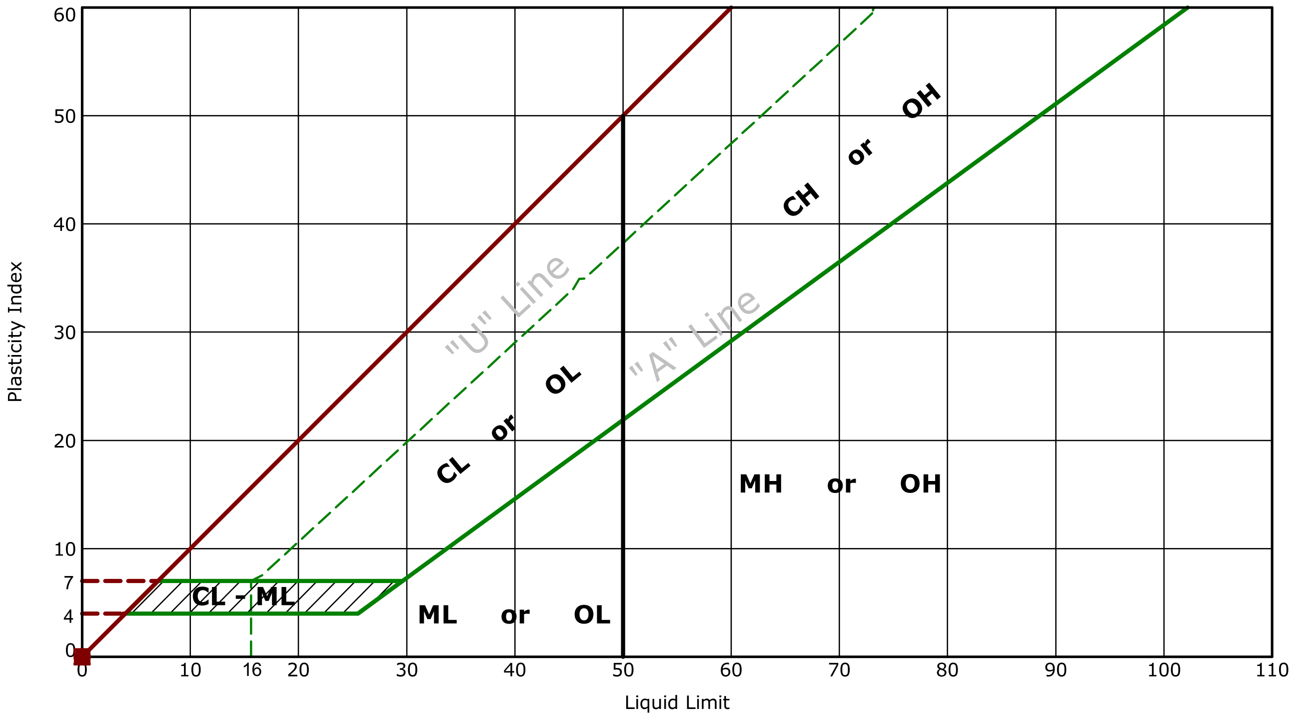
Boring Started
10-30-2023

Boring Completed
10-30-2023

1505 Old Happy Jack Rd
Cheyenne, WY

Atterberg Limit Results

ASTM D4318

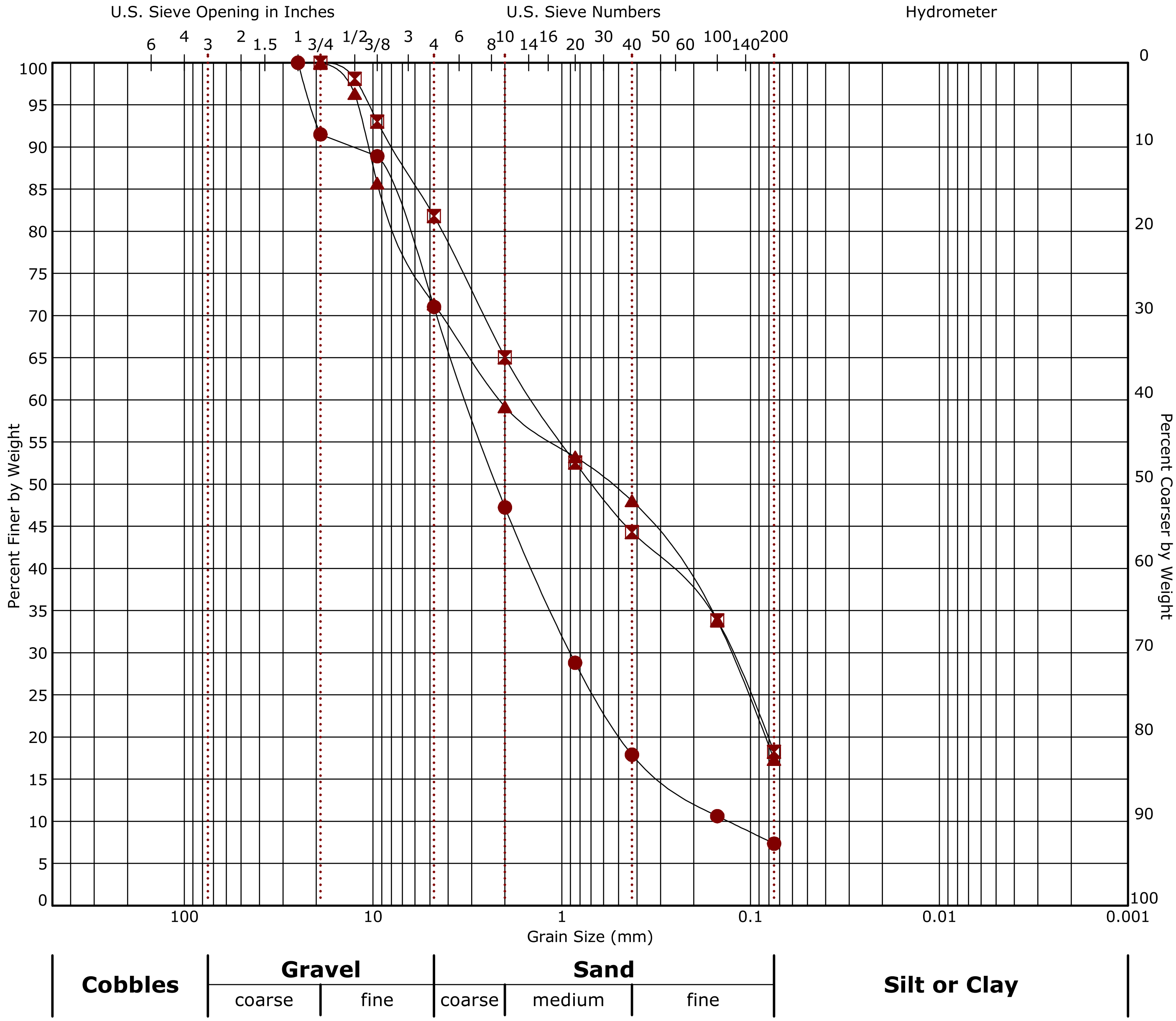


	Boring ID	Depth (Ft)	LL	PL	PI	Fines	USCS	Description
●	1	2 - 4	NP	NP	NP	7.4	SW-SM	WELL-GRADED SAND with SILT and GRAVEL
⊠	1	4 - 6	NP	NP	NP	18.3	SM	SILTY SAND with GRAVEL
▲	2	6 - 7.5	NP	NP	NP	17.3	SM	SILTY SAND with GRAVEL

1505 Old Happy Jack Rd
Cheyenne, WY

Grain Size Distribution

ASTM D422 / ASTM C136



	Boring ID	Depth (Ft)	USCS Classification	USCS	AASHTO	LL	PL	PI	Cc	Cu
●	1	2 - 4	WELL-GRADED SAND with SILT and GRAVEL	SW-SM	A-1-a (0)	NP	NP	NP	1.93	24.18
⊠	1	4 - 6	SILTY SAND with GRAVEL	SM	A-1-b (0)	NP	NP	NP		
▲	2	6 - 7.5	SILTY SAND with GRAVEL	SM	A-1-b (0)	NP	NP	NP		

	Boring ID	Depth (Ft)	D_{100}	D_{60}	D_{30}	D_{10}	%Cobbles	%Gravel	%Sand	%Fines	%Silt	%Clay
●	1	2 - 4	25	3.18	0.898	0.132	0.0	29.0	63.7	7.4		
⊠	1	4 - 6	19	1.414	0.126		0.0	18.2	63.5	18.3		
▲	2	6 - 7.5	19	2.121	0.128		0.0	28.6	54.0	17.3		

CHEMICAL LABORATORY TEST REPORT

Project Number: 24221110
Service Date: 7/18/2023
Report Date: 7/18/2023

Terracon
1505 Old Happy Jack Road
Cheyenne, Wyoming 82001
(307) 632-9224

Project: Cheyenne VAMC EHRM Infrastructure Upgrades

Sample Location	B-2
Sample Depth (ft.)	1-6
pH Analysis, ASTM - G51	7.4
Water Soluble Sulfate (SO4), ASTM C1580 (mg/kg)	Nil
Sulfides, ASTM - D4658 (mg/kg)	0.4
Chlorides, ASTM - D512 (mg/kg)	26
RedOx, ASTM - D1498 (mV)	+214
Total Salts, ASTM - D1125 (mg/kg)	2,847
Resistivity, ASTM - G187 (ohm-cm)	4,550

Reviewed By: Steve Anderson
Laboratory Manager

Supporting Information

Contents:

Note: All attachments are one page unless noted above.

Terracon
1505 Old Happy Jack Rd
Cheyenne, WY

General Notes

Sampling	Water Level	Field Tests
Auger Cuttings Modified California Ring Sampler Standard Penetration Test	Water Initially Encountered Water Level After a Specified Period of Time Water Level After a Specified Period of Time Cave In Encountered Water levels indicated on the soil boring logs are the levels measured in the borehole at the times indicated. Groundwater level variations will occur over time. In low permeability soils, accurate determination of groundwater levels is not possible with short term water level observations.	N Standard Penetration Test Resistance (Blows/Ft.) (HP) Hand Penetrometer (T) Torvane (DCP) Dynamic Cone Penetrometer UC Unconfined Compressive Strength (PID) Photo-Ionization Detector (OVA) Organic Vapor Analyzer

Descriptive Soil Classification

Soil classification as noted on the soil boring logs is based Unified Soil Classification System. Where sufficient laboratory data exist to classify the soils consistent with ASTM D2487 "Classification of Soils for Engineering Purposes" this procedure is used. ASTM D2488 "Description and Identification of Soils (Visual-Manual Procedure)" is also used to classify the soils, particularly where insufficient laboratory data exist to classify the soils in accordance with ASTM D2487. In addition to USCS classification, coarse grained soils are classified on the basis of their in-place relative density, and fine-grained soils are classified on the basis of their consistency. See "Strength Terms" table below for details. The ASTM standards noted above are for reference to methodology in general. In some cases, variations to methods are applied as a result of local practice or professional judgment.

Location And Elevation Notes

Exploration point locations as shown on the Exploration Plan and as noted on the soil boring logs in the form of Latitude and Longitude are approximate. See Exploration and Testing Procedures in the report for the methods used to locate the exploration points for this project. Surface elevation data annotated with +/- indicates that no actual topographical survey was conducted to confirm the surface elevation. Instead, the surface elevation was approximately determined from topographic maps of the area.

Strength Terms

Relative Density of Coarse-Grained Soils (More than 50% retained on No. 200 sieve.) Density determined by Standard Penetration Resistance			**Consistency of Fine-Grained Soils** (50% or more passing the No. 200 sieve.) Consistency determined by laboratory shear strength testing, field visual-manual procedures or standard penetration resistance			
Relative Density	**Standard Penetration or N-Value (Blows/Ft.)**	**Ring Sampler (Blows/Ft.)**	**Consistency**	**Unconfined Compressive Strength Qu (psf)**	**Standard Penetration or N-Value (Blows/Ft.)**	**Ring Sampler (Blows/Ft.)**
Very Loose	0 - 3	0 - 5	Very Soft	less than 500	0 - 1	< 3
Loose	4 - 9	6 - 14	Soft	500 to 1,000	2 - 4	3 - 5
Medium Dense	10 - 29	15 - 46	Medium Stiff	1,000 to 2,000	4 - 8	6 - 10
Dense	30 - 50	47 - 79	Stiff	2,000 to 4,000	8 - 15	11 - 18
Very Dense	> 50	> 80	Very Stiff	4,000 to 8,000	15 - 30	19 - 36
			Hard	> 8,000	> 30	> 36

Relevance of Exploration and Laboratory Test Results

Exploration/field results and/or laboratory test data contained within this document are intended for application to the project as described in this document. Use of such exploration/field results and/or laboratory test data should not be used independently of this document.

Unified Soil Classification System

Criteria for Assigning Group Symbols and Group Names Using Laboratory Tests [A]				Soil Classification	
				Group Symbol	**Group Name** [B]
Coarse-Grained Soils: More than 50% retained on No. 200 sieve	**Gravels:** More than 50% of coarse fraction retained on No. 4 sieve	**Clean Gravels:** Less than 5% fines [C]	Cu≥4 and 1≤Cc≤3 [E]	GW	Well-graded gravel [F]
			Cu<4 and/or [Cc<1 or Cc>3.0] [E]	GP	Poorly graded gravel [F]
		Gravels with Fines: More than 12% fines [C]	Fines classify as ML or MH	GM	Silty gravel [F, G, H]
			Fines classify as CL or CH	GC	Clayey gravel [F, G, H]
	Sands: 50% or more of coarse fraction passes No. 4 sieve	**Clean Sands:** Less than 5% fines [D]	Cu≥6 and 1≤Cc≤3 [E]	SW	Well-graded sand [I]
			Cu<6 and/or [Cc<1 or Cc>3.0] [E]	SP	Poorly graded sand [I]
		Sands with Fines: More than 12% fines [D]	Fines classify as ML or MH	SM	Silty sand [G, H, I]
			Fines classify as CL or CH	SC	Clayey sand [G, H, I]
Fine-Grained Soils: 50% or more passes the No. 200 sieve	**Silts and Clays:** Liquid limit less than 50	**Inorganic:**	PI > 7 and plots above "A" line [J]	CL	Lean clay [K, L, M]
			PI < 4 or plots below "A" line [J]	ML	Silt [K, L, M]
		Organic:	$\frac{LL\ oven\ dried}{LL\ not\ dried} < 0.75$	OL	Organic clay [K, L, M, N]
					Organic silt [K, L, M, O]
	Silts and Clays: Liquid limit 50 or more	**Inorganic:**	PI plots on or above "A" line	CH	Fat clay [K, L, M]
			PI plots below "A" line	MH	Elastic silt [K, L, M]
		Organic:	$\frac{LL\ oven\ dried}{LL\ not\ dried} < 0.75$	OH	Organic clay [K, L, M, P]
					Organic silt [K, L, M, Q]
Highly organic soils:	Primarily organic matter, dark in color, and organic odor			PT	Peat

[A] Based on the material passing the 3-inch (75-mm) sieve.
[B] If field sample contained cobbles or boulders, or both, add "with cobbles or boulders, or both" to group name.
[C] Gravels with 5 to 12% fines require dual symbols: GW-GM well-graded gravel with silt, GW-GC well-graded gravel with clay, GP-GM poorly graded gravel with silt, GP-GC poorly graded gravel with clay.
[D] Sands with 5 to 12% fines require dual symbols: SW-SM well-graded sand with silt, SW-SC well-graded sand with clay, SP-SM poorly graded sand with silt, SP-SC poorly graded sand with clay.
[E] $Cu = D_{60}/D_{10} \qquad Cc = \frac{(D_{30})^2}{D_{10} \times D_{60}}$
[F] If soil contains ≥ 15% sand, add "with sand" to group name.
[G] If fines classify as CL-ML, use dual symbol GC-GM, or SC-SM.
[H] If fines are organic, add "with organic fines" to group name.
[I] If soil contains ≥ 15% gravel, add "with gravel" to group name.
[J] If Atterberg limits plot in shaded area, soil is a CL-ML, silty clay.
[K] If soil contains 15 to 29% plus No. 200, add "with sand" or "with gravel," whichever is predominant.
[L] If soil contains ≥ 30% plus No. 200 predominantly sand, add "sandy" to group name.
[M] If soil contains ≥ 30% plus No. 200, predominantly gravel, add "gravelly" to group name.
[N] PI ≥ 4 and plots on or above "A" line.
[O] PI < 4 or plots below "A" line.
[P] PI plots on or above "A" line.
[Q] PI plots below "A" line.